



WT: VDDML & 2DML

Q1. (ai) Claim: if g & h agree on free variables, then $V_{M,g}(\phi, u) = V_{M,h}(\phi, u)$, for each $u \in U$

If. Suppose g & h agree on free variables

We prove that $V_{M,g}(\phi, u) = V_{M,h}(\phi, u)$ for each u on the complexity of ϕ by induction on the complexity of ϕ

BC: $\phi = \prod \alpha_1, \dots, \alpha_n$

Then, each term α_i is either a constant or a freely occurring variable

If α_i is a constant, then for any pair g, h

$$[\alpha_i]_g = [\alpha_i]_h$$

If α_i is a variable, by assumption

$$[\alpha_i]_g = [\alpha_i]_h$$

Then, for any world $u \in U$,

$$\langle [\alpha_1]_g, \dots, [\alpha_n]_g, u \rangle = \langle [\alpha_1]_h, \dots, [\alpha_n]_h, u \rangle$$

$$\therefore \langle [\alpha_1]_g, \dots, [\alpha_n]_g, u \rangle \in I(\prod^n) \text{ iff}$$

$$\langle [\alpha_1]_h, \dots, [\alpha_n]_h, u \rangle \in I(\prod^n)$$

$$\therefore V_{M,g}(\phi, u) = V_{M,h}(\phi, u)$$

$$\text{IH: } \forall n < m, V_{M,g}(\phi_n, u) = V_{M,h}(\phi_n, u)$$

Then, we have four cases for ϕ_m

$$1) \phi_m = \sim \phi_n$$

That's not an iff, By IH, $V_g(\phi_n, u) = V_h(\phi_n, u)$

Make $V_g(\phi_m, u) = V_h(\phi_m, u)$ iff

$$V_g(\sim \phi_n, u) = V_h(\sim \phi_n, u)$$

$$\text{iff } V_g(\phi_n, u) = V_h(\phi_n, u)$$

$$2) \phi_m = \phi_{n_1} \rightarrow \phi_{n_2}, \text{ where } n_1, n_2 < m$$

$$\text{By IH, } V_g(\phi_{n_1}, u) = V_h(\phi_{n_1}, u)$$

$$\text{and } V_g(\phi_{n_2}, u) = V_h(\phi_{n_2}, u)$$

$$\text{iff } V_g(\phi_{n_1} \rightarrow \phi_{n_2}, u) = V_h(\phi_{n_1} \rightarrow \phi_{n_2}, u)$$

What are ϕ_n and ϕ_m ?
if you have an indexing
you need to define it

and the
other different

$$\text{iff } V_g(\phi_m, u) = V_h(\phi_m, u)$$

$$3) \phi_m = \bigvee_{\alpha} \phi_n$$

$$\text{By IH, } V_g(\phi_n, u) = V_h(\phi_n, u)$$

$$V_g(\bigvee_{\alpha} \phi_n, u) = 1 \text{ iff } \forall u \in D, V_{g_u}^{\alpha}(\phi_n, u) = 1$$

$$\text{iff } \forall u \in D, V_{h_u}^{\alpha}(\phi_n, u) = 1 \text{ iff } V_h(\bigvee_{\alpha} \phi_n, u) = 1$$

where $g_u^{\alpha}, h_u^{\alpha}$ agree on free variables

$$\therefore V_g(\phi_m, u) = V_h(\phi_m, u)$$

$$4) \phi_m = \Box \phi_n$$

$$\text{By IH, } V_g(\phi_n, u) = V_h(\phi_n, u)$$

for all $u \in U$

$$\text{Then, } V_g(\Box \phi_n, u) = 1 \text{ iff } \forall v \in U \text{ s.t. } R_{uv}$$

$$V_g(\phi_n, u) = 1 \text{ iff } \forall v \in U \text{ s.t. } R_{uv}$$

$$V_h(\phi_n, u) = 1 \text{ iff } V_h(\Box \phi_n, u) = 1$$

$$\therefore V_g(\phi_m, u) = V_h(\phi_m, u) = 0$$

ii) (aim: if $[\alpha]_{m,g} = [\beta]_{m,g}$, then $V_{m,g}(\phi, u) = V_{m,g}(\phi(\beta/\alpha), u)$, for each $u \in W$)

Pf. Suppose $[\alpha]_{m,g} = [\beta]_{m,g}$. We prove that $V_{m,g}(\phi, u) = V_{m,g}(\phi(\beta/\alpha), u)$ for each $u \in W$ by induction on ϕ 's complexity.

BC: WLOG, $\phi = \pi \alpha_1, \dots, \alpha_n$ (it does not matter how many α s there are or what α is)

Then by assumption, for any world u

$$\langle [\alpha]_{m,g}, [\alpha_1]_{m,g}, \dots, [\alpha_n]_{m,g} \rangle \in I_u(\pi)$$

$$\text{iff } \langle [\beta]_{m,g}, [\alpha_1]_{m,g}, \dots, [\alpha_n]_{m,g} \rangle \in I_u(\pi)$$

$$\Rightarrow \langle [\alpha]_{m,g}, [\alpha_1]_{m,g}, \dots, [\alpha_n]_{m,g}, u \rangle \in I(\pi)$$

$$\text{iff } \langle [\beta]_{m,g}, [\alpha_1]_{m,g}, \dots, [\alpha_n]_{m,g}, u \rangle \in I(\pi)$$

$$\Rightarrow V_{m,g}(\phi, u) = V_{m,g}(\phi(\beta/\alpha), u)$$

$$\text{IH: } \forall n < m, V_{m,g}(\phi_n, u) = V_{m,g}(\phi_n(\beta/\alpha), u)$$

4 Cases to consider:

$$1) \phi_n = \sim \phi_n. \text{ Then, } V_{m,g}(\phi_n, u) = V_{m,g}(\phi_n(\beta/\alpha), u) \quad (\text{IH})$$

$$\text{iff } V_{m,g}(\sim \phi_n, u) = V_{m,g}(\sim \phi_n(\beta/\alpha), u)$$

$$\text{iff } V_{m,g}(\phi_n, u) = V_{m,g}(\phi_n(\beta/\alpha), u)$$

$$2) \phi_n = \phi_{n_1} \rightarrow \phi_{n_2}, \quad n_1, n_2 < m$$

$$\text{Then, } V_{m,g}(\phi_n, u) = V_{m,g}(\phi_{n_2}(\beta/\alpha), u) \quad \&$$

$$V_{m,g}(\phi_{n_1}, u) = V_{m,g}(\phi_{n_1}(\beta/\alpha), u) \text{ iff } V_{m,g}(\phi_{n_1} \rightarrow \phi_{n_2}, u) = V_{m,g}((\phi_{n_1} \rightarrow \phi_{n_2})(\beta/\alpha), u),$$

$$\text{iff } V_{M,g}(\phi_n, u) = V_{M,g}(\phi_n(\beta/\alpha), u)$$

3) $\phi_n = \forall \alpha \phi_n$. $\rightarrow$ $\forall \alpha$ need to do a case distinction here of the variable being quantified over, it could be α , β or γ some other

doesn't necessarily follow $\leftarrow$

$$\therefore \text{for all } u \in D, \quad V_{g_u}^\alpha(\phi_n, u)$$

$$= V_{g_u}^\alpha(\phi_n(\beta/\alpha), u)$$

$$\text{if } [\alpha]_{M,g} = [\beta]_{M,g} \therefore V_g(\forall \alpha \phi_n, u) = V_g(\forall \alpha \phi_n(\beta/\alpha), u)$$

$$\text{then } V_g(F_\alpha) = V_g(F_\beta) \therefore V_{M,g}(\phi_n, u) = V_{M,g}(\phi_n(\beta/\alpha), u)$$

$$= V_g(\forall \alpha F_\beta)$$

$$4) \phi_n = \Box \phi_n$$

but that may be different from $V_g(\forall \alpha F_\beta)$

$$\text{By IH, } V_{M,g}(\phi_n, u) = V_{M,g}(\phi_n(\beta/\alpha), u) \text{ for each } u \in W$$

So, at any world $v \in W$ s.t. R_{vv} ,

$$V_{M,g}(\phi_n, v) = V_{M,g}(\phi_n(\beta/\alpha), v) \quad \checkmark$$

$$\therefore V_{M,g}(\phi_n, v) = 1 \quad \forall v \in W \text{ s.t. } R_{vv}$$

$$\text{iff } V_{M,g}(\phi_n(\beta/\alpha), v) = 1 \quad \forall v \in W \text{ s.t. } R_{vv}$$

$$\therefore V_{M,g}(\Box \phi_n, w) = V_{M,g}(\Box \phi_n(\beta/\alpha), w)$$

$$\therefore V_{M,g}(\phi_n, w) = V_{M,g}(\phi_n(\beta/\alpha), w)$$

Claim proven

b) i. Claim: $\not\models_{\text{forall}} \forall \alpha \phi \rightarrow \phi(\beta/\alpha)$

Countermodel: $W = \{u\}$ $R = \emptyset$

$D = \{u, v\}$

$D_u = \{u\}$, let $\phi = f_\alpha$

$I(F) = \{\langle u, u \rangle\}$



thus, g could be such that $\langle [B]_{M,g}, u \rangle = \langle v, u \rangle$

so, $\langle [B]_{M,g}, u \rangle \notin I(F)$

while $\forall d \in D_u, V_{g_d}^\alpha(F, u) = 1$

$\therefore$ there is some g s.t. $V_g(\forall \alpha f_\alpha, u) = 1$

while $V_g(F_\beta, u) = 0$

ii. Claim: $\models_{\text{forall}} \forall \alpha \phi \rightarrow (\exists \gamma (\gamma = \beta) \rightarrow \phi(\beta/\alpha))$ with $\gamma \neq \alpha, \beta$

Pf. AFC that there is an M s.t.

$V_{M,g}(\forall \alpha \phi \rightarrow (\exists \gamma (\gamma = \beta) \rightarrow \phi(\beta/\alpha)), u) = 0$

$\therefore V_{M,g}(\forall \alpha \phi, u) = 1, V_{M,g}(\exists \gamma (\gamma = \beta), u) = 1,$

①

⑤

$V_{M,g}(\phi(\beta/\alpha), u) = 0$

②

From ①: $\forall d \in D_u, V_{g_d}^\alpha(\phi, u) = 1$

From ②: $\exists d \in D_u$ s.t. $[\gamma]_{g_d}^{\beta/\alpha} = [\beta]_{g_d}^{\beta/\alpha}$ $\xrightarrow{d \mapsto d}$ These d's need to be distinct

From ③: $V_{M,g_d}(\phi(\beta/\alpha), u) = 0 \rightarrow$ How do you get this?

but this is a contradiction since ② implies

$$\text{that } [\beta]_{g_d^d} \in D_\omega$$

and ① implies that for that d which β denotes

$$V_{g_d^d}^\alpha(\phi, u) = 1$$

$$\text{since } \langle [\alpha]_{g_d^d} \rangle = \langle [\beta]_{g_d^d} \rangle$$

$$V_{g_d^d}^{\alpha\beta}(\phi(\beta/\alpha), u) = 1$$

✓

$$\Rightarrow V_{g_d^d}(\phi(\beta/\alpha), u) = 1 \quad \checkmark$$

iii. Claim: $\models_{\text{model}} \forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi)$ where α does not occur free in ϕ

Pf. AFC that $V_{M,g}(\forall \alpha (\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \forall \alpha \psi), u) = 0$

$$\therefore V_{M,g}(\forall \alpha (\phi \rightarrow \psi), u) = 1 \quad \&$$

①

$$V_{M,g}(\phi \rightarrow \forall \alpha \psi, u) = 0$$

②

$$\text{from ②: } \forall d \in D_u, V_{g_d^d}^\alpha(\phi \rightarrow \psi, u) = 1$$

$$\therefore V_{g_d^d}^\alpha(\psi, u) = 1 \text{ or } V_{g_d^d}^\alpha(\phi, u) = 0$$

$$\text{from ②: } V_g(\phi, u) = 1, \text{ for some } d' \in D_u, V_{g_{d'}^{d'}}(\psi, u) = 0$$

Since g & $g_{d'}^{d'}$ agree on free variables as α does not occur free in ϕ ,
 $V_g(\phi, u) = V_{g_{d'}^{d'}}(\phi, u) = 1$

Since $V_{g_{d'}}^{\alpha}(\phi, u) = 1$, it must be the case

that $V_{g_{d'}}^{\alpha}(\psi, u) = 1$ due to ① ✓

but for that $d' \in D$, $V_{g_{d'}}^{\alpha}(\psi, u) = 0$ ✗

Q2 (a) Let F be increasing mean that R_{uu} implies $D_u \subseteq D_u$

Claim: $\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$ is valid on a frame F iff
 F is increasing

($\Leftarrow$) Suppose F is increasing i.e.

if R_{uu} , then $D_u \subseteq D_u$

for reductio, $V_{M, g}(\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi, u) = 0$

$\therefore V_{M, g}(\Box \forall \alpha \phi, u) = 1, V_{M, g}(\forall \alpha \Box \phi, u) = 0$

$\Rightarrow \forall u \in W$ s.t. $R_{uu}, \quad \Rightarrow \exists d' \in D_u$ s.t. $V_{g_{d'}}^{\alpha}(\Box \phi, u) = 0$
 $V_{M, g}(\forall \alpha \phi, u) = 1 \quad \Rightarrow$ for some world $w' \in W$ s.t. $R_{uw'}$

$\Rightarrow \forall d \in D_u, V_{g_d}^{\alpha}(\phi, u) = 1 \quad V_{g_{d'}}^{\alpha}(\phi, u') = 0$

but this is a contradiction
increasing means that $D_u \subseteq D_{u'}$ and
let us pick some D_u s.t. $D_{u'} \subseteq D_u$ & $D_u \subseteq D_{u'}$, not
 $\Rightarrow$ this always holds since at minimum we can always pick this
 $D_u = D_{u'}$ (due to F being increasing)

then, given $D_{u'} \subseteq D_u$, we can choose $d = d'$

$$\text{s.t. } V_{g_{d'}}^{\alpha}(\phi, u) = 1$$

$$\text{but } V_{g_{d'}}^{\alpha}(\phi, u) = 0 \quad \times$$

($\Rightarrow$) Suppose that $\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$ is valid on a frame F

then, suppose for reductio that F is not increasing

i.e. there is a pair, $u \not\leq u'$ s.t. Ruu' but $D_u \not\subseteq D_{u'}$

Then, we can create a countermodel

$$V_{M,g}(\Box \forall \alpha \phi, u) = 1 \quad \& \quad V_{M,g}(\forall \alpha \Box \phi, u) = 0$$

$$\Rightarrow V_{M,g}(\forall \alpha \phi, u) = 1 \quad \Rightarrow V_{g_{d'}}^{\alpha}(\Box \phi, u) = 0 \text{ for some } d' \in D_u$$

$$\forall u \in U \text{ s.t. } Ruu'$$

$$\Rightarrow V_{g_{d'}}^{\alpha}(\phi, u) = 0 \text{ for some } u \in U$$

$$\Rightarrow V_{g_d}^{\alpha}(\phi, u) = 1$$

$$\text{s.t. } Ruu'$$

$$\forall d \in D_u$$

$$U = \{u_1, u_2\}$$

$$R = \{\langle u_1, u_2 \rangle\}$$

$$D_{u_1} = \{u, v\}$$

$$D_{u_2} = \{u\}$$

$$I(F) = \{\langle u \rangle\}$$

$$\text{so, let } [\alpha]_{g_{d'}}^{\alpha} = v$$

contradicting the assumption. $\therefore F$ is increasing

$$D = \bigcup_{i \geq 1} D_{u_i}$$

$$\text{let } \phi = F_{\alpha}$$

$$I_{u_1}(F) = \{\langle u \rangle\}$$

$$\text{so, let } [\alpha]_{g_{d'}}^{\alpha} = v$$

(b) Claim: if all instances of $\Box \forall x \phi \rightarrow \forall x \Box \phi$ & $\Diamond \Box \phi \rightarrow \phi$ are valid on a frame F

then F is locally constant i.e. $R_{uu} \Rightarrow D_u = D_w$

Lemma: if $\Diamond \Box \phi \rightarrow \phi$ is valid on a frame F , $R_{uu} \Rightarrow R_{uw}$ i.e.

pf. Assume that there is no M s.t. R is symmetric of W

$$V_{M,g}(\Diamond \Box \phi \rightarrow \phi, u) = 0$$

$$\text{Then, } V_{M,g}(\Diamond \Box \phi, u) = 1, V_{M,g}(\phi, u) = 0$$

$$\text{then, } V_{M,g}(\Box \phi, u) = 1$$

for some $u \in W$ s.t. R_{uu}

$$\text{then, } V_{M,g}(\phi, v) = 1$$

Does not this
step assume
symmetry
already? or
how do you get
that R_{uw} ?

for all v s.t. R_{uv}

then $V_{M,g}(\phi, u) = 1$ for contradiction/impossibility

but this means that u must be accessible from u

so, R_{uu}

$$\therefore R_{uu} \Rightarrow R_{uw}$$

Main pf. Using the lemma, we reduce the claim to:

Claim : if all instances of $\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$ is valid on F & R is symmetric on W
then F is locally constant

From 2a) if all instances of $\Box \forall \alpha \phi \rightarrow \forall \alpha \Box \phi$ is valid on F , then F is increasing i.e. $R_{uw} \Rightarrow D_u \subseteq D_w$

By symmetry, this means that R_{wu} as well
so $D_w \subseteq D_u$

$\therefore$ if $D_u \subseteq D_w$ & $D_w \subseteq D_u$

then $D_w = D_u$ ✓

i.e. F is locally constant QED

Q3. a) i. Claim: $\models_{2D} \phi \leftrightarrow D @ \phi$

Pf. AFC $V_{M,g}^2(\phi \leftrightarrow D @ \phi, u, u) = 0$

$$\Rightarrow V_{M,g}^2(\phi, u, u) \neq V_{M,g}^2(D @ \phi, u, u)$$

Case 1: Suppose $V_{M,g}^2(\phi, u, u) = 0, V_{M,g}^2(D @ \phi, u, u) = 1$

$$\text{then } V_{M,g}^2(@ \phi, w, w') = 1 \text{ for all } w' \in W$$

$$\text{then } V_{M,g}^2(\phi, w, w) = 1 \quad \times$$

Case 2: Suppose $V_{M,g}^2(\phi, u, u) = 1, V_{M,g}^2(D @ \phi, u, u) = 0$

$$\text{then } V_{M,g}^2(@ \phi, u, u') = 0 \text{ for some } u' \in W$$

$$\text{then } V_{M,g}^2(\phi, u, u) = 0 \quad \times$$

$$\text{So } \models_{2D} \phi \leftrightarrow D @ \phi$$

ii. Claim: $\not\models_{G,D} \phi \leftrightarrow D @ \phi$

$$D = \{u, v\} \quad V_{M,g}(\phi, u_1, u_2) = 0$$

$$W = \{u_1, u_2\} \quad V_{M,g}(\phi, u_1, u_1) = 1$$

$$I_{u_2}(F) = \{\langle v \rangle\} \quad \phi = F_a$$

$$I_{u_1}(F) = \{\langle u \rangle\} \quad I(a) = u$$

(b) i. Claim: $\not\models_{2D} \Box(\phi \leftrightarrow \Box \Box \phi)$

$$W = \{w_1, w_2\}$$

$$D = \{u\} \quad I(a) = u$$

$$I_{u_1}(F) = \{\langle u \rangle\}$$

$$I_{u_2}(F) = \emptyset$$

$$\therefore V_{M,g}(\Box(\phi \leftrightarrow \Box \Box \phi), w_1, w_1) = 0$$

$$\text{since } V_{M,g}(\phi \leftrightarrow \Box \Box \phi, w_1, w_2) = 0$$

$$\text{as } V_{M,g}(\phi, w_1, w_2) = 0$$

$$\text{while } V_{M,g}(\Box \Box \phi, w_1, w_2) = 1$$

✓

$$\Leftrightarrow V_{M,g}(\Box \phi, w_1, w_1) = 1 \quad \text{✗}$$

$$V_{M,g}(\Box \phi, w_1, w_2) = 1$$

$$\text{since } V_{M,g}(\phi, w_1, w_1) = 1$$

ii. Claim: $\not\models_{GD} \Box(\phi \leftrightarrow \Box \Box \phi)$

Pf. Use the same model as in bi)

✓

$$\text{i.e. } W = \{u_1, u_2\}$$

$$V = \{u\}$$

$$I_{u_1}(F) = \{\langle u \rangle\}, I_{u_2}(F) = \emptyset$$

$$I(a) = u$$

(c) i. Claim: $\models_{2D} \Box X(\phi \leftrightarrow \Box \Box \phi)$

$$\text{Pf. AFC } V_{M,g}^2(\Box X(\phi \leftrightarrow \Box \Box \phi), u, v) = 0$$

$\therefore$ for some $v \in W$,

$$V_{M,g}^2(X(\phi \leftrightarrow \Box \Box \phi), u, v) = 0$$

$$V_{M,g}^2(\phi \leftrightarrow \Box \Box \phi, v, v) = 0$$

$$V_{M,g}^2(\phi, v, v) \neq V_{M,g}^2(\Box \Box \phi, u, v)$$

then we can show the exact same contradiction obtains as in ai) ✗

ii. Claim: $\models_{G2D} \Box X(\phi \leftrightarrow \Box \Box \phi)$

$$\text{Pf. AFC } V_{M,g}(\Box X(\phi \leftrightarrow \Box \Box \phi), v, u) = 0$$

$$\text{for some } u', V_{M,g}(X(\phi \leftrightarrow \Box \Box \phi), v, u') = 0$$

$$\Rightarrow V_{M,g}(\phi \leftrightarrow \Box \Box \phi, u', u') = 0$$

and the exact same contradiction as in ai) occurs ✗

Q4. (a) Some non-actual thing could exist

$\exists x (\sim @F_x \wedge \Diamond F_x)$ 'there is an x such that x does not actually exist and it is possible that x exists'

$\Diamond \exists x (\sim @F_x \wedge F_x)$ 'It is possible that there is some x that does not actually exist but exists'

1) Scope of $\Diamond$ or possibility is ambiguous

(b) $\Diamond \forall x (\sim @F_x \rightarrow F_x) \Rightarrow$ all non-actual things, possibly existing together

$\forall x (\sim @F_x \rightarrow \Diamond F_x) \Rightarrow$ all non-actual things could individually exist

(c) $\Box X (\Diamond \forall x (@R_x \rightarrow R_x) \wedge \forall y (@P_x \rightarrow R_x)) \Rightarrow$ swap

$\Box X (\Diamond \forall x (@R_x \rightarrow P_x) \wedge \Diamond \forall y (@P_x \rightarrow R_x)) \Rightarrow$ collectively change

$\Box X (\forall x (@R_x \rightarrow \Diamond P_x) \wedge \forall y (@P_x \rightarrow \Diamond R_x)) \Rightarrow$ individually change

(d) $\sim \exists x I_x \vee \exists x (I_x \wedge \forall y (@I_y \rightarrow y=x) \wedge @I_x)$

Assignment 1)

(a) σ_{Eng} : If something is actually valuable, then it is valuable ✓

(b) i) $\models_{2D} \forall x (@F_x \rightarrow F_x)$

$$\text{AFC that } V_{M,g}^2(\forall x (@F_x \rightarrow F_x), u, u) = 0$$

$$\therefore V_{M,g_{d'}}^2(@F_x \rightarrow F_x, u, u) = 0 \text{ for some } d' \in D$$

$$\therefore V_{M,g_{d'}}^2(@F_x, u, u) = 1 \Rightarrow V_{M,g_{d'}}^2(F_x, u, u) = 1 \quad \checkmark$$

$$\therefore V_{M,g_{d'}}^2(F_x, u, u) = 0 \quad \times$$

$$\text{since both } [x]_{g_{d'}}^x \notin I(F) \text{ \& } [x]_{g_{d'}}^z \in I(F)$$

ii) $\not\models_{2D} \Box \forall x (@F_x \rightarrow F_x)$

$$V_{M,g}(\Box \forall x (@F_x \rightarrow F_x), u, u) = 0$$

$$\therefore V_{M,g}(\forall x (@F_x \rightarrow F_x), u, v) = 0 \text{ for some } v \in W$$

$$\therefore V_{M,g_d}^2(@F_x \rightarrow F_x, u, v) = 0 \text{ for some } d \in D$$

$$\therefore V_{M,g_d}^2(@F_x, u, v) = 1 \quad \therefore V_{M,g_d}^2(F_x, u, v) = 0$$

$$\Rightarrow V_{M,g_d}^2(F_x, u, u) = 1$$

$$W = \{u, u_2\}$$

$$D = \{d\}$$

$$I(F) = \{\langle d, u \rangle\} \quad \checkmark$$

$$(iii) \models_{\mathcal{D}} F @ \forall x (@ F_x \rightarrow F_x) \quad V_{M,g}^2(F\phi, v, w) = 1 \text{ iff } \forall v' \in W, V_{M,g}^2(\phi, v, v') = 1$$

$$\text{AFC } V_{M,g}(F @ \forall x (@ F_x \rightarrow F_x), w, w) = 0$$

$$\Rightarrow V_{M,g}(@ \forall x (@ F_x \rightarrow F_x), v, w) = 0 \text{ for some } v \in W$$

$$\Rightarrow V_{M,g}(\forall x (@ F_x \rightarrow F_x), v, v) = 0$$

$$\Rightarrow V_{M,g_d}(@ F_x \rightarrow F_x, v, v) = 0 \text{ for some } d \in D$$

✓

$$\Rightarrow V_{M,g_d}^2(@ F_x, v, v) = 1, \quad \Rightarrow V_{M,g_d}^2(F_x, v, v) = 0$$

$$\therefore V_{M,g_d}^2(F_x, v, v) = 1 \quad \text{Then } \langle [x]_{g_d} \rangle \notin I(F) \text{ \&}$$

$$\langle [x]_{g_d}^x \rangle \notin I(F) \text{ \&}$$